

INVITED TALK / ABSTRACT

Stability of extremal domains

Marcos Petrucio Cavalcante

Universidade Federal de Alagoas

Abstract

In this talk, we discuss the notion of stable extremal domains for the first Dirichlet eigenvalue of the Laplace-Beltrami operator on Riemannian manifolds. We classify stable extremal domains in the 2-sphere and in higher-dimensional spheres with minimal boundary. We also establish topological bounds for stable domains in compact Riemannian surfaces, assuming either nonnegative total Gaussian curvature or sufficiently small area. This is joint work with Ivaldo Nunes.

Keywords: stable extremal domains; first Dirichlet eigenvalue; Laplace-Beltrami operator; Riemannian manifolds; 2-sphere; higher-dimensional spheres; minimal boundary; topological bounds; Gaussian curvature